Loan Delinquency & NPA Monitor: Project Write-up

Excel project using Power Query, Tables, PivotTables, PivotCharts and Slicers. Data: Prosper Loan Data (public Kaggle sample of US loans).

1. Objective

Build an Excel dashboard that works like a collections MIS report at a lending company. It answers:

- How many active loans are overdue, and how late are they (DPD buckets)?
- How much money sits in overdue loans and in NPAs (90+ days overdue)?
- Which borrower segments (rating, loan size, state, year) carry the most risk?
- Can a manager filter all of this in one click? (slicers)

2. Data

- Source: Prosper Loan Data on Kaggle. 113,937 rows and 81 columns, loans from Nov 2005 to Mar 2014, in US dollars. Public sample data, not company data.
- After cleaning: 113,066 loans in one Excel table (`tblloans`) with 18 columns (11 kept, 1 added in Power Query, 6 helper columns).

3. What we did and how

Stage 1: Clean the data (Power Query)

Data > Get Data > From Text/CSV > Transform Data. The query steps:

1. Loaded the CSV, promoted the first row to headers, and set data types.
2. Kept 11 of the 81 columns: Term, LoanStatus, ProsperRating (Alpha), BorrowerState, LoanKey, LoanCurrentDaysDelinquent, LoanMonthsSinceOrigination, LoanOriginalAmount, LoanOriginationDate, MonthlyLoanPayment, LP_CustomerPayments.
3. Removed duplicates on `LoanKey` (871 repeated loans).

4. Replaced blank BorrowerState with "Unknown" and blank ProsperRating with "Pre-2009".
5. Added a conditional column `Loan_Book`: Chargedoff or Defaulted = Written-off; Completed or Cancelled = Closed; everything else = Active.
6. Close & Load into an Excel table.

Check: 113,066 loans = Active 57,990 + Closed 38,066 + Written-off 17,010.

Stage 2: Helper columns (table formulas)

Each formula was typed once in a new column of the table and Excel filled the whole column (a calculated column). References like `[@Loan_Book]` point to the same row.

- `DPD_Bucket` (which overdue band): `=IF([@Loan_Book]<>"Active","Not Active",IFS([@LoanCurrentDaysDelinquent]=0,"Current",[@LoanCurrentDaysDelinquent]<=30,"1-30",[@LoanCurrentDaysDelinquent]<=60,"31-60",[@LoanCurrentDaysDelinquent]<=90,"61-90",TRUE,"90+"))`. IFS returns the answer for the first true condition. `TRUE` at the end is the catch-all for anything above 90 days.
- `Overdue_Flag` (1 if an active loan is late): `=IF(AND([@Loan_Book]="Active",[@LoanCurrentDaysDelinquent]>0),1,0)`
- `NPA_Flag` (1 if an active loan is 90+ days late): `=IF(AND([@Loan_Book]="Active",[@LoanCurrentDaysDelinquent]>90),1,0)`
- `WrittenOff_Flag`: `=IF([@Loan_Book]="Written-Off",1,0)`
- `Size_Band` (loan size group): `=IFS([@LoanOriginalAmount]<5000,"Under 5K",[@LoanOriginalAmount]<10000,"5K-10K",[@LoanOriginalAmount]<20000,"10K-20K",TRUE,"20K+")`
- `Orig_Year`: `=YEAR([@LoanOriginationDate])`

Key trick: the **Average** of a 0/1 flag in a pivot is a percentage. Average of `Overdue_Flag` is the delinquency rate, and the Sum is the number of loans.

Stage 3: Analysis (12 PivotTables on the `Calc` sheet)

Every pivot has `Loan_Book` in Filters: Active for the risk views, Closed + Written-off for the write-off view.

- DPD bucket summary: loan count and amount per bucket
- Delinquency rate by ProsperRating

- Delinquency and NPA rate by Size_Band
- Top 10 states by overdue loans (Top 10 filter)
- Write-off rate by rating (finished loans)
- Delinquency rate and loan count by origination year
- Two KPI pivots (active book, write-off rate)
- Four small pivots feeding the charts (buckets, rating, size, state)

Techniques: Value Field Settings (custom names, percentage formats), manual item ordering, pivot drill-down (double-click a value), Refresh All.

Stage 4: Dashboard

- **Title and 5 KPI cards** (rectangles plus text boxes). Each number box is linked to a cell that holds a `GETPIVOTDATA` formula, so the cards react to the slicers:
  - Active Loans: `=GETPIVOTDATA("Count of LoanKey",$B$79)` gives 57,990
  - Exposure: `=GETPIVOTDATA("Sum of LoanOriginalAmount",$B$79)` gives $595.29M
  - Delinquency Rate: `=GETPIVOTDATA("Average of Overdue_Flag",$B$79)` gives 3.55%
  - NPA Loans: `=GETPIVOTDATA("Sum of NPA_Flag",$B$79)` gives 320
  - Written-Off Rate: `=GETPIVOTDATA("WrittenOff_Flag",$B$86)` gives 15%
- **4 PivotCharts**: overdue loans by DPD bucket (column), delinquency rate by rating (column), delinquency rate by loan size (column), top 10 states by overdue loans (bar). Field buttons hidden, titles in text boxes.
- **3 slicers**: ProsperRating, Size_Band, Orig_Year, connected to every pivot through Report Connections.
- Gridlines off, one colour theme, custom number formats (two commas at the end of a format show millions).

4. Formulas and tools used

Area	What we used
Formulas	IF, IFS, AND, YEAR, GETPIVOTDATA, table references like `[@Column]`, cell links for text boxes

Area	What we used
Power Query	Get Data (CSV), Promote Headers, Change Type, Choose Columns, Remove Duplicates, Replace Values, Conditional Column, Close & Load, Refresh All
Tables	Ctrl+T, named table, calculated columns
PivotTables	Filters, Rows, Values, Count/Sum/Average, Value Field Settings, Top 10 filter, manual sorting, drill-down
PivotCharts	Column and bar charts linked to pivots
Slicers	3 slicers, Report Connections
Dashboard design	Shapes, text boxes linked to cells, custom number formats, gridlines off
Shortcuts	Alt N V (PivotTable), Alt F1 (chart), Ctrl+T, Ctrl+Z, Ctrl+Space, Alt+F5 and Ctrl+Alt+F5 (refresh)

5. Difficulties and how we solved them

Problem	Cause	Fix
Duplicate loans	871 repeated LoanKeys	Remove Duplicates on LoanKey
Huge "days late" on dead loans	Charged-off and defaulted loans carry up to 2,704 days, which would fill the 90+ bucket	Split loans into Active / Closed / Written-off, and bucketed Active only
29,084 loans with no rating	The rating system only started in 2009	Labelled them "Pre-2009" instead of deleting them
Blank row showing 100% in the rating pivot	One rating cell in the table had been changed by hand	Found it with pivot drill-down, restored it with Refresh All. Lesson: never edit a Power Query table by hand

Problem	Cause	Fix
Zeros in the write-off and overdue pivots	Wrong Loan_Book filter (each flag is 1 only inside its own group)	Set the right filter on each pivot
Refresh security warning	Excel is cautious about external data connections	Allowed the refresh for our own file
Wrong pivot order	Text sorts A to Z (A before AA, size bands alphabetical)	Dragged items into logical order
Unreadable chart	55,933 current loans dwarf the overdue buckets, and multi-measure pivots plot every value	Used one-measure pivots and hid the Current bucket from the chart

6. Key findings

Active loan book unless stated. Sample data.

- **Portfolio:** 113,066 loans. 57,990 active ($595.3M original amount), 38,066 closed, 17,010 written off (15.0% of all loans, 30.9% of loans that have finished).
- **Delinquency:** 2,057 active loans are overdue, which is 3.55%. 96.45% are current. Overdue loans hold about $17.0M (2.9% of active exposure).
- **DPD buckets:** 1-30 days: 1,065 loans. 31-60: 361. 61-90: 311. 90+ (NPA): 320 loans, which is 0.55% of loans and $2.57M.
- **Rating drives risk:** delinquency climbs steadily from AA 0.71% to A 1.43%, B 2.31%, C 3.38%, D 5.49%, E 6.35% and HR 9.02%. That is about 13 times higher at the bottom.
- **Same pattern in write-offs:** among finished loans, the write-off rate runs from 4.7% (AA) to 38.2% (HR). Today's delinquency by rating is an early warning of future losses.
- **Smaller loans are riskier:** Under 5K 5.2%, 5K-10K 3.8%, 10K-20K 2.8%, 20K+ 1.6%.
- **Where overdue loans are:** CA 206, NY 150, FL 146, TX 107, GA 96. These are counts, so they mostly follow where the most loans are.
- **Vintage trap:** 2011 loans 8.2%, 2012 7.6%, 2013 2.8%, 2014 0.15%. Newer loans look safe only because they have had less time to go overdue.

7. Suggestions

For the business

- Chase the 1-30 day bucket first (1,065 loans). It is the cheapest to recover and stops loans rolling into 31-60.
- Watch the 31-60 and 61-90 buckets (672 loans) closely, since they are next in line to become NPAs.
- Tighten underwriting and monitoring for D, E and HR ratings, and for small-ticket loans.
- Compare loan cohorts at the same age before saying a newer vintage is safer.

To improve the project

- The Written-Off Rate card shows 15%, which is written-off loans out of all loans. The finished-loans figure is 30.9%. Either relabel the card "% of all loans" or filter that pivot to Closed + Written-off.
- `Loan_Book` has two spellings ("Written-Off" and "Written-off"). Excel ignores the difference, so the flags are right, but use one spelling in Power Query.
- Exposure is the original loan amount, not the balance still owed, because the data has no loan-level outstanding balance. Keep the label "Exposure (original amount)".
- Possible extensions: a monthly trend, roll-rate (how loans move between buckets), and a Power BI version.